SHILPA - INCLUSIVE MOBILE LEARNING APP FOR VISUALLY, HEARING, PHYSICALLY AND COGNITIVELY IMPAIRED STUDENTS (GRADE 3-5)

Thanthirige K.S.G.I.

IT22177650

BSc (Hons) in Information Technology Specializing in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

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DECLARATION OF THE CANDIDATES AND SUPERVISOR

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Name	**Student ID**	**Signature**	**Date**
Thanthirige K.S.G.I.	IT22177650		26/4/2026
Wijesooriya C.D.	IT22078728		26/4/2026
Perera G.W.R.S.	IT22064622		26/4/2026
Mathota Arachchi S.S.	IT22070876		26/4/2026

The above candidates have carried out research for the bachelor's degree Dissertation under our supervision.

Role	**Name**	**Signature**	**Date**
Supervisor	Prof. Nuwan Kodagoda		
Co-Supervisor	Mrs. Jenny Krishara		

ABSTRACT

Shilpa is presented as an inclusive mobile learning application designed to support primary education for Sri Lankan students in Grades 3–5 with visual, hearing, physical, and cognitive impairments within a single unified platform. Existing educational technologies present clear limitations because accessibility features are typically implemented in isolation, resulting in fragmented learning experiences that exclude learners with different impairment types. To address this, an integrated Flutter-based application was developed with four cooperating modules: a Braille-driven assessment workflow that converts quizzes to Braille-ready PDFs and uses a Convolutional Neural Network (CNN) to recognise scanned answer sheets for visually impaired learners; a Sinhala Sign Language (SSL) lesson and gesture-imitation module that uses MediaPipe hand-landmark extraction and a Bidirectional Long Short-Term Memory (BiLSTM) network for hearing-impaired learners; an adaptive multimodal interaction framework combining adaptive dwell-time selection, gaze tracking, voice control, and a hybrid voice-plus-gaze mode for physically impaired learners; and a TensorFlow Lite cognitive classification module that personalises gamified micro-activities to a student's curriculum level. The system was evaluated through model performance analysis and field user testing at the National Institute of Education, the School for the Deaf in Tangalle, and partner primary schools. The CNN Braille recogniser reached 88–92 % accuracy, the BiLSTM SSL recogniser 96–98 % validation accuracy, the hybrid voice-plus-gaze mode achieved 94 % selection accuracy with the lowest task time, and the cognitive classifier achieved 82–83 % accuracy on real student data. The evaluation confirms that integrating multiple accessibility modules within a single Sinhala-medium platform enhances usability, supports independent learning, and reduces accessibility barriers for inclusive primary education in the Sri Lankan context.

Keywords: inclusive education, mobile learning, Sinhala Sign Language, Braille recognition, multimodal interaction, adaptive learning

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
ASL	American Sign Language
BiLSTM	Bidirectional Long Short-Term Memory
BSL	British Sign Language
CL	Curriculum Level
CNN	Convolutional Neural Network
EMA	Exponential Moving Average
HCI	Human–Computer Interaction
IT	Information Technology
ITS	Intelligent Tutoring System
JWT	JSON Web Token
LSTM	Long Short-Term Memory
ML	Machine Learning
MLM	Machine Learning Model
NIE	National Institute of Education
OCR	Optical Character Recognition
PDF	Portable Document Format
REST	Representational State Transfer
SEND	Special Educational Needs and Disabilities
SLIIT	Sri Lanka Institute of Information Technology
SLSL	Sinhala Sign Language
SSL	Sri Lankan Sign Language
STT	Speech-to-Text
SVM	Support Vector Machine
TFLite	TensorFlow Lite
TLS	Transport Layer Security
TTS	Text-to-Speech
UDL	Universal Design for Learning
UI	User Interface
UX	User Experience
WCAG	Web Content Accessibility Guidelines

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1. INTRODUCTION

Shilpa is an inclusive mobile learning application developed for Sri Lankan primary school students in Grades 3–5 who experience visual, hearing, physical, or cognitive learning difficulties. The platform combines four accessibility modules — voice and Braille support for visually impaired learners, Sinhala Sign Language (SSL) lesson delivery and gesture practice for hearing-impaired learners, multimodal touch-free interaction for physically impaired learners, and adaptive curriculum-level personalization for learners with cognitive and learning difficulties — into a single Flutter-based application. Existing educational technologies typically address only a single disability type, follow international rather than local sign languages, and rarely provide structured assessment workflows that work without sighted assistance. Shilpa was developed to address these limitations by integrating multiple accessibility methods such as voice interaction, sign-language recognition, gesture-based control, and adaptive learning strategies in one localized Sinhala-medium platform aligned with the Sri Lankan primary curriculum.

1.1 Background and Literature Review

1.1.1 Visual impairment support

Visual impairment significantly affects how students access, understand, and interact with educational content. Most teaching methods rely heavily on printed text, images, diagrams, and graphical interfaces, which create serious barriers for students who are blind or have low vision. In Sri Lanka, although some applications provide basic Text-to-Speech (TTS) or screen reader support, they are limited to passive listening and do not effectively support active learning tasks such as answering questions, practising exercises, or receiving feedback independently. A particularly important limitation is the lack of proper assessment methods for visually impaired learners in digital systems, because most quizzes are designed for visual interaction and require touch or typed responses.

Several research efforts have explored assistive technologies in this space. The CU Eye Reader integrates Optical Character Recognition (OCR) with TTS to let visually impaired users read printed text by ear, achieving good accuracy for general reading but providing no structured educational features such as lessons or assessments. The KNFB Reader offers similar OCR-with-speech support and is fast and accurate, but it is not child-oriented and lacks structured learning. Sánchez and colleagues showed that audio-based virtual environments and 3D spatial audio cues improve orientation and concept understanding for visually impaired users, supporting the use of directional audio in lessons. More recent work demonstrates that deep learning can recognise Braille patterns directly from images: BrailleSegNet shows segmentation methods for Braille datasets, Fly-LeNet converts Braille images into multilingual text, and CNN-based recognition has been shown to handle Braille characters with high accuracy. However, these recognition pipelines are not connected to learning workflows, leaving a clear gap between Braille writing practice and automated assessment.

1.1.2 Hearing impairment support

Hearing impairment is one of the most common challenges affecting primary school students in Sri Lanka, where most lessons depend on spoken instruction. International solutions designed for American Sign Language (ASL) or British Sign Language (BSL) cannot be transferred directly to Sinhala Sign Language (SSL), because the gestures, grammar, and cultural context differ. Sign Language Recognition (SLR) systems typically combine gesture capture using MediaPipe or BlazePose with sequence models such as Long Short-Term Memory (LSTM), Bidirectional LSTM (BiLSTM), or Gated Recurrent Units (GRUs) to interpret dynamic gestures.

Vision-based ASL recognition with MediaPipe and SVM classification has reached approximately 98–99 % accuracy for static signs, and Leap Motion combined with Deep Neural Networks has reached approximately 89 % for selected SSL letters and words. However, sensor-based solutions are expensive and reduce accessibility, while recognition-only systems do not include lessons, gesture practice, or measurable feedback. Local efforts including the SSL Learning System, Hastha, BeMyVoice, EasyTalk, HANDTALK, IHear, Visual Kids, and the Tamil Sign Language smart system have built static-sign detectors, translation engines, or e-learning platforms, but most stop at recognition or simple multiple-choice quizzes and do not provide curriculum-aligned imitation activities with measurable scoring for primary learners. SLRNet shows that real-time LSTM-based ASL recognition is viable on commodity devices, while pose-estimation studies highlight that skeletal representations alone can lose important information during hand-to-hand and hand-to-face interactions, motivating sequence-based dynamic-gesture models for SSL.

1.1.3 Physical impairment support

Students with physical disabilities in Grades 3–5 often face significant challenges in navigating and interacting with traditional touch-based mobile applications. Existing assistive technologies are typically designed for adults and focus on a single input modality such as switch access, eye gaze, head pointing, eyelid gestures, or tilt input. Borgestig, Hsieh, Wilson, Andreassen, Masayko, and Hemmingsson have demonstrated, across multiple longitudinal and intervention studies, that gaze-based assistive technology can support independence and daily participation for children with severe motor impairments, but their work is confined to gaze as a single modality and does not extend to multimodal classroom learning.

Other strands of work focus on isolated alternative inputs: Cicek and colleagues explore head-based pointing on smartphones, Fan and colleagues study eyelid gestures, Castellucci and MacKenzie evaluate tilt-based text entry, and Wobbrock and Myers examine trackball entry for motor-impaired users. Mott and colleagues report on smartphone photography accessibility, and Malu and colleagues explore smartwatch interactions for upper-body motor impairments. Chan and colleagues show that eye-tracking training can improve memory and learning, while Rytterström and colleagues capture teachers' experiences with gaze-controlled learning tools. Local Sri Lankan work by Jayasekara and colleagues has produced a smart wheelchair with mobile-app support but does not target inclusive primary learning. Across this body of work, alternative input methods remain siloed, child-friendly safety controls are rare, and very few systems combine multiple modalities into a single child-centered learning platform.

1.1.4 Cognitive impairment support

Cognitive disabilities affect children's ability to learn, remember, and apply knowledge. Approximately 2–3 % of school-aged children in developing countries live with cognitive disabilities, and within Sri Lanka the standardized education system offers limited flexibility for individualized learning, with fewer than 5 % of children who require support able to access it. Adaptive learning, gamified lessons, and visual aids have been shown to enhance learning outcomes for these learners, and AI-driven adaptive learning platforms can adjust lesson difficulty in real time based on performance and cognitive profile.

Studies on adaptive learning systems for intellectual disabilities, AI-driven personalised learning pathways, machine learning approaches for cognitive load optimisation, game-based micro-learning, gamification in special education, animated pedagogical agents, Universal Design for Learning (UDL) on mobile devices, and AI-based psychometric tutoring frameworks have each contributed important ideas. However, most existing tools are designed for English-speaking or Western users, are not aligned with the Sri Lankan curriculum, and do not combine curriculum-level adaptation with Sinhala-language support, animated emotional guidance, gamified rewards, and offline-friendly delivery. Studies of cultural localization in South Asian educational technology and of the digital divide in Sri Lankan special education confirm the importance of Sinhala-first design and low-resource-friendly deployment.

1.2 Research Gap

Across the four impairment areas, the literature reveals a consistent gap. Existing systems address either a single disability or a single interaction modality, depend on specialised hardware, or focus on translation or recognition rather than structured learning. None of the surveyed systems combines all four impairment types in a single Sinhala-medium platform aligned with the Sri Lankan primary curriculum, and none integrates Braille-based assessment, SSL imitation games, multimodal touch-free interaction, and adaptive cognitive-level personalisation under one application.

1.2.1 Visual impairment gap

Existing assistive solutions for visually impaired learners are dominated by screen readers and basic TTS tools that support reading but not active learning. There is no system that allows students to answer quizzes using printed Braille, upload their responses, and receive automated evaluation. Stereo and 3D spatial audio cues are used in navigation and entertainment but are rarely applied to school lessons. Braille-Ready Document Generation (BRDG) tools exist as standalone utilities but are not integrated into learning platforms. Multiple feedback channels — vibration alerts, contextual audio prompts, screen-reader compatibility, biometric login, TTS and STT — are typically implemented separately rather than as part of a single unified system.

1.2.2 Hearing impairment gap

Most hearing-impaired learning systems focus on recognition or translation, use ASL or BSL rather than SSL, rely on specialised hardware, or stop at simple multiple-choice quizzes without measurable feedback. There is no single platform that combines SSL video lessons, real-time camera-based sign imitation, gesture scoring tied to a curriculum, gamified lesson activities, and longitudinal progress reporting for primary-school learners on commodity Android hardware.

1.2.3 Physical impairment gap

Studies on dwell time, gaze, head pointing, eyelid gestures, and tilt input validate each modality individually, but rarely integrate them. Few systems combine adaptive dwell-time activation, gaze stabilisation with calibration, voice command parsing, and a hybrid voice-plus-gaze fusion mode within a single child-friendly educational application. Safety features such as emergency stop, accidental-touch cancellation, and visual countdowns, together with bilingual (Sinhala and English) command support, are rarely addressed in the literature.

1.2.4 Cognitive impairment gap

Generic adaptive-learning platforms exist, but no solution combines curriculum-level adaptation, Sinhala-language delivery, animated emotional guidance, gamified micro-activities aligned with the Sri Lankan primary syllabus, on-device classification, and offline-friendly operation in one platform tailored for Sri Lankan learners.

1.3 Research Problem

Despite advances in assistive technology, primary school students with visual, hearing, physical, or cognitive impairments in Sri Lanka continue to face significant barriers when accessing digital learning. Existing tools address one disability at a time, are typically built for English-speaking contexts, lack alignment with the Sri Lankan primary curriculum, and rarely support active assessment for the impairment categories they target. The central research problem of this project is therefore:

How can a single Sinhala-medium mobile learning platform integrate Braille-based assessment for visually impaired learners, SSL gesture-imitation learning for hearing-impaired learners, multimodal touch-free interaction for physically impaired learners, and adaptive curriculum-level personalisation for learners with cognitive and learning difficulties, while remaining usable on commodity Android hardware and aligned with the Grade 3–5 syllabus?

Sub-problems addressed by each module are summarised below.

1.3.1 Visual impairment sub-problem

Most assistive tools support content reading but not the complete cycle of lesson, interaction, practice, and assessment. There is no system that lets young visually impaired learners answer quizzes using printed Braille and receive immediate automated feedback that does not require sighted assistance, particularly in Sinhala-medium classrooms.

1.3.2 Hearing impairment sub-problem

Tools using ASL or BSL do not transfer to SSL, and existing Sinhala-based applications mostly provide video lessons or subtitles without measurable real-time feedback on signing accuracy. Young learners need interactive lessons aligned with the local curriculum, with gesture scoring, gamified imitation activities, and progress reports for parents and teachers.

1.3.3 Physical impairment sub-problem

Most assistive tools rely on a single input method — gaze, dwell time, voice, or switch — and force users to depend on that method even when it is fatiguing. Voice systems are trained on adult speech and perform poorly for children. Local-language commands, child-friendly safety features, and personalised settings such as dwell-time duration and button size are typically absent.

1.3.4 Cognitive impairment sub-problem

Standardised teaching methods often overwhelm slower learners while failing to challenge faster learners, and existing international tools do not combine Sinhala-language support, curriculum-level adaptation, gamified micro-activities, and emotional guidance. Without such localisation, children with cognitive and learning difficulties continue to fall behind.

1.4 Research Objectives

1.4.1 Main Objective

To design, develop, and evaluate Shilpa — an inclusive Sinhala-medium mobile learning application — that integrates Braille-based assessment, Sinhala Sign Language gesture-imitation learning, multimodal touch-free interaction, and adaptive curriculum-level personalisation in a single Flutter platform aligned with the Sri Lankan Grade 3–5 syllabus.

1.4.2 Specific Objectives

1. Implement full voice-based interaction for navigation, lesson delivery, and quizzes using Text-to-Speech (TTS) and Speech-to-Text (STT) within the Flutter environment, and integrate stereo and 3D spatial audio for concept understanding.
2. Generate Braille-ready PDF question papers, capture Braille answer sheets via image upload, and use a trained Convolutional Neural Network (CNN) to recognise Braille patterns and deliver automated audio feedback.
3. Deliver curriculum-aligned SSL video lessons with Sinhala subtitles, supported by animated visual explanations and a Sinhala chatbot for question answering.

4. Implement camera-based MediaPipe hand-landmark extraction and a Bidirectional Long Short-Term Memory (BiLSTM) gesture-recognition model deployed on-device via TensorFlow Lite, together with a gesture-scoring algorithm and sign-imitation games.
5. Implement adaptive dwell-time selection, gaze and head-pose navigation with stability filtering and personalised calibration, voice-command navigation in Sinhala and English with fuzzy matching, and a hybrid voice-plus-gaze fusion mode.
6. Implement an on-device TensorFlow Lite cognitive classifier that uses a 28-feature vector derived from gamified assessment activities to predict each learner's curriculum level (below, average, above) and personalise micro-activities accordingly.
7. Provide progress-tracking dashboards, reward and badge systems, and progress reports for parents and teachers, with secure authentication and anonymised data storage.
8. Evaluate the integrated system through model performance analysis and field user testing at the National Institute of Education, the School for the Deaf in Tangalle, and partner primary schools, and report results across all four modules.

2. METHODOLOGY

This chapter describes the methodology adopted to design, develop, and evaluate the four modules of Shilpa. It is organised across the sections prescribed by the SLIIT dissertation guideline: requirements gathering and analysis, feasibility study, tools and technologies, per-module methodology, system architecture, project requirements, aspects of the system, commercialization aspects of the product, and testing and implementation.

2.1 Requirements Gathering and Analysis

Requirements were gathered through a combination of literature review, structured field visits to schools and education institutions, stakeholder interviews, and prototype demonstrations. Field visits and data-collection sessions were conducted at the National Institute of Education (NIE) Maharagama, the NIE Braille Press, the School for the Deaf in Sitinamaluwa, Beliatta and Tangalle, the Zonal Education Office in Hatton, and Kalugala Maha Vidyalaya. Across these visits the team observed how visually impaired students read and write Braille, how deaf students used Sinhala Sign Language in classroom routines, how students with motor difficulties interacted with mobile devices, and how teachers identified students with cognitive and learning difficulties.

All gathered inputs were systematically analysed and categorised into functional and non-functional requirements. Use cases were modelled for each user role — student, teacher, and parent — and requirements were prioritised using MoSCoW analysis (Must-have, Should-have, Could-have, Won't-have). The core learning, recognition, interaction, and progress-tracking features were treated as mandatory deliverables. Sinhala-language interface, child-friendly visuals, and offline-friendly operation emerged as cross-cutting requirements applicable to all four modules.

2.2 Feasibility Study

A four-dimension feasibility study confirmed that the project could be delivered within the constraints of an undergraduate research project.

2.2.1 Technical feasibility

All core technical requirements — Braille image recognition, hand-landmark extraction with sequence classification, real-time gaze estimation and voice control, and on-device cognitive classification — are supported by mature open-source tools. Flutter provides a cross-platform mobile UI; TensorFlow/Keras with TensorFlow Lite supports model training and on-device inference; MediaPipe Hands provides 21-keypoint landmark extraction; Google ML Kit Face Detection supports gaze tracking; OpenCV supports preprocessing for Braille images; and the speech_to_text package supports continuous Sinhala-locale recognition. The Node.js/Express backend and MongoDB database provide flexible storage. All technologies are well documented and have strong community support.

2.2.2 Operational feasibility

Stakeholder engagement at the partner schools confirmed strong willingness to adopt a Sinhala-medium digital learning tool provided it is easy to use and accompanied by a brief orientation session. Students demonstrated positive engagement during prototype demonstrations across all four modules. The system requires no specialised hardware beyond a standard Android smartphone.

2.2.3 Economic feasibility

Development cost is kept low through entirely free and open-source tooling. Firebase free-tier services and the Google Play Store distribution channel are sufficient for the pilot deployment phase, and no licensing fees are required for any component of the technical stack.

2.2.4 Schedule feasibility

Development was structured across iterative Agile sprints, each producing a testable increment of the four modules. The most critical features — Braille recognition, SSL gesture recognition, multimodal interaction, and cognitive classification — were delivered first, with refinements applied in later sprints based on testing feedback.

2.3 Tools, Technologies and Algorithms

The Shilpa platform is built on a shared cross-module technology stack summarised in Table 2.1, with module-specific algorithms detailed in the per-module methodology sections.

Layer	**Technology / Algorithm**	**Purpose**
Frontend	Flutter (Dart)	Cross-platform mobile UI
Backend	Node.js / Express (TypeScript)	REST API gateway, JWT auth, multipart upload
Database	MongoDB (with Firebase / AWS S3 for assets)	User profiles, progress, lesson assets
ML Service	Python FastAPI	Hosts CNN, BiLSTM, and cognitive classifier
Visual recogniser	CNN (TensorFlow / Keras)	Braille digit recognition
Hearing recogniser	MediaPipe Hands + BiLSTM	Dynamic SSL gesture recognition
Physical interaction	Google ML Kit Face Detection, speech_to_text	Gaze, head-pose, blink, voice
Cognitive recogniser	TensorFlow Lite classifier	On-device curriculum-level prediction
Speech / audio	flutter_tts, speech_to_text, audioplayers	TTS, STT, sound effects
PDF / Braille	pdfkit, jsPDF, OpenCV preprocessing	Braille-ready PDF, Braille image preprocessing
Auth / hosting	Firebase Auth, AWS S3, GitHub	Auth, asset hosting, version control

Table 2.1: Cross-module technology and algorithm stack.

2.4 Visual Impairment Support — Methodology

The visual impairment module enables a fully voice-driven learning experience for blind and low-vision students. Students listen to lessons using TTS and provide responses using STT, with vibration feedback and screen-reader compatibility for ease of use.

2.4.1 Voice-based interaction

Voice-based navigation is implemented in Flutter using flutter_tts for audio output and speech_to_text for input. Recognised commands such as navigation, selection, and answer entry are mapped to predefined actions, and contextual audio prompts and vibration feedback guide the student through lessons. Stereo and 3D spatial audio cues are used in lesson narration to improve concept understanding.

2.4.2 Braille-ready document generation

Quiz questions are automatically converted into a Braille-ready PDF document using a server-side generator that outputs structured Braille formatting compatible with the NIE Braille Press conventions. Parents or teachers print the PDF and the student answers using a slate and stylus in the traditional Braille writing workflow.

2.4.3 CNN-based Braille recognition

A custom Braille digit dataset was manually collected and resized to a consistent format. Image preprocessing applies grayscale conversion, thresholding, and per-cell segmentation to highlight raised dots and remove background noise. A Convolutional Neural Network with multiple convolutional and pooling layers extracts spatial features, a flatten layer feeds dense layers, a dropout layer reduces overfitting, and a Softmax output layer assigns probabilities to each Braille digit class. The trained model is integrated into the mobile pipeline so that when an answer-sheet image is uploaded, the system preprocesses, segments, and classifies each cell, compares predicted digits with the correct answers, and delivers the result through audio feedback.

2.5 Hearing Impairment Support — Methodology

The hearing impairment module delivers SSL video lessons with Sinhala subtitles and integrates real-time camera-based sign imitation with gesture scoring.

2.5.1 Lesson delivery and gamified learning

Lessons are delivered as SSL videos with Sinhala subtitles and supporting diagrams, ensuring all key concepts are understandable without audio. A Sinhala chatbot, AI-generated post-lesson quizzes, sign-imitation games, and animated story-based explanations make the learning experience engaging for primary-school learners. Progress dashboards expose lesson completion and quiz history to teachers and parents.

2.5.2 Hand-landmark extraction with MediaPipe

MediaPipe Hands extracts 21 hand keypoints per video frame, returning normalised (x, y, z) triplets. The 63 numerical values per frame are stacked across a fixed window into a [frames × 63] sequence array, which forms the structured input to the gesture recognition model and replaces raw pixel-level processing.

2.5.3 BiLSTM gesture recognition

A Bidirectional Long Short-Term Memory (BiLSTM) network implemented in TensorFlow / Keras processes the landmark sequence in both forward and backward temporal directions. Two stacked BiLSTM layers (128 and 64 units) with dropout (0.3) and a Softmax output were trained with Adam optimizer, categorical cross-entropy loss, batch size 32, and up to 80 epochs with early stopping on an 80/20 stratified split. The model is exported to TensorFlow Lite for efficient on-device inference.

2.5.4 Gesture scoring

When a student performs a sign, the system computes the cosine similarity between the student's landmark sequence and a stored reference sequence for the target gesture, producing an accuracy score between 0 and 100. A personalised calibration step on first use adjusts thresholds for individual differences in hand size, signing speed, and range of motion.

2.6 Physical Impairment Support — Methodology

The physical impairment module is implemented as an Adaptive Multimodal Interaction Optimization Framework with five interaction modalities — Standard Touch, Adaptive Dwell-to-Touch, Head-Gaze Tracking with Blink-to-Select, Voice-Based Control, and a Hybrid Multimodal Fusion mode — that users can switch between at runtime via a floating action button. A centralised InteractionStatusService acts as a global event bus enabling cross-modal communication.

2.6.1 Adaptive dwell-time selection

The AdaptiveDwellService dynamically adjusts the dwell activation duration between 1.0 and 3.0 seconds, starting at 2.0 seconds. The algorithm uses a reinforcement-style asymmetric feedback loop: after five consecutive successful selections the dwell duration is decreased by 0.1 s; after three consecutive cancellations the dwell duration is increased by 0.2 s. The asymmetry intentionally favours stability over speed. Adapted durations are persisted using SharedPreferences so the system remembers each user's optimal dwell time across sessions, and a broadcast Stream notifies the UI to display real-time feedback.

2.6.2 Eye and head-gaze tracking with blink-to-select

The EyeTrackingService uses Google ML Kit Face Detection to extract facial landmarks and Euler angles (yaw, pitch) at approximately 30 FPS. Each frame produces a GazeData object containing the estimated screen position, stability status, and blink probability. Stage 1 maps yaw and pitch to screen coordinates relative to the centre with a configurable sensitivity factor. Stage 2 rejects any frame where the cursor jumps more than 150 px from the previous position. Stage 3 applies Exponential Moving Average smoothing with $\alpha = 0.25$. Stage 4 keeps a rolling 15-frame history and classifies the gaze as stable when variance falls below a personalised threshold. A 5-point calibration computes a drift offset and a personalised stability threshold (average variance × 1.5, clamped between 15.0 and 60.0 px). Blink-to-select uses ML Kit's eye-open probabilities with a double-blink confirmation protocol within 2.5 s to prevent accidental selection.

2.6.3 Voice-based command navigation

The SpeechService wraps the speech_to_text package into a continuous listening loop with automatic restart and Sinhala (si-LK) locale priority. An acoustic confidence threshold ($\geq$ 0.5) filters low-confidence recognitions. The VoiceCommandParser implements a multi-strategy NLP pipeline: numeric ID extraction supporting Arabic digits and Sinhala phonetic numbers, bilingual keyword matching with Levenshtein-distance fuzzy fallback at 75 % similarity, and a semantic-label fallback that matches the spoken text against on-screen button labels. The VoiceFocusService maintains a runtime registry of all visible interactive elements so commands such as "say one" or "say eka" select the first registered button on the current screen.

2.6.4 Hybrid multimodal fusion

The InputAwareButton is a polymorphic UI component that adapts to the active InputMode, supporting all five modalities simultaneously in Hybrid mode. In Hybrid mode, gaze acts as a pointer and voice acts as the selection trigger, combining the precision of gaze with the intentionality of voice while a centralised controller prevents conflicts between inputs.

2.7 Cognitive Impairment Support — Methodology

The cognitive support module is designed as a closed-loop adaptive system that integrates game-based assessment, machine-learning-based classification, and personalised activity delivery.

2.7.1 Game-based data collection

When a student logs in and accesses the curriculum-level (CL) assessment module, three timed interactive games — shape matching, colour matching, and bubble popping — capture detailed interaction-based performance data: total touches, valid taps, correct and incorrect responses, hint usage, response time, and missed targets. These metrics reflect the student's attention, accuracy, and response behaviour in a natural and engaging environment.

2.7.2 Feature engineering

Raw data is processed to derive accuracy, inefficiency, hint rate, and miss rate. Combined with the raw inputs, these form a fixed 28-feature vector. A consistent feature order is enforced across training and inference.

2.7.3 Classification model

A supervised neural classifier built with TensorFlow / Keras categorises students into below-average, average, and above-average curriculum levels. The architecture uses an input normalisation layer, two hidden Dense layers (128 units each) with batch normalisation, ReLU activation, and dropout, and a final softmax output with three neurons. Training uses Adam optimisation, sparse categorical cross-entropy, validation split monitoring, and class weighting to handle imbalance. The dataset is split 90 % labelled (further split into training and testing) and 10 % unseen for prediction.

2.7.4 Deployment and adaptive personalisation

The trained model is converted to TensorFlow Lite and deployed for on-device inference. At runtime the 28-feature vector is passed to the model, which outputs the predicted CL with class probabilities and a confidence score. The result is sent to a Node.js backend, validated, and stored in MongoDB to maintain a historical record per student. When the student accesses the cognitive dashboard, the latest prediction is retrieved and used to personalise subsequent activities: below-average → foundational activities, average → intermediate activities, above-average → advanced activities, creating a continuous adaptive cycle.

2.8 System Architecture

Shilpa was designed as a unified Flutter-based mobile application backed by a Node.js / Express REST API and a Python FastAPI ML inference service. MongoDB stores user profiles, progress, and lesson metadata; AWS S3 hosts SSL lesson video assets and large model artefacts; Firebase Authentication provides role-based login for students, teachers, and parents. Each of the four accessibility modules shares the same authentication, progress-tracking, and reporting infrastructure, and routes requests to the appropriate ML pipeline (CNN for Braille, BiLSTM for SSL, on-device TensorFlow Lite for cognitive classification). The physical impairment module operates almost entirely on-device using MediaPipe / ML Kit. The system overview is shown in Figure 1 and the high-level architecture in Figure 2.

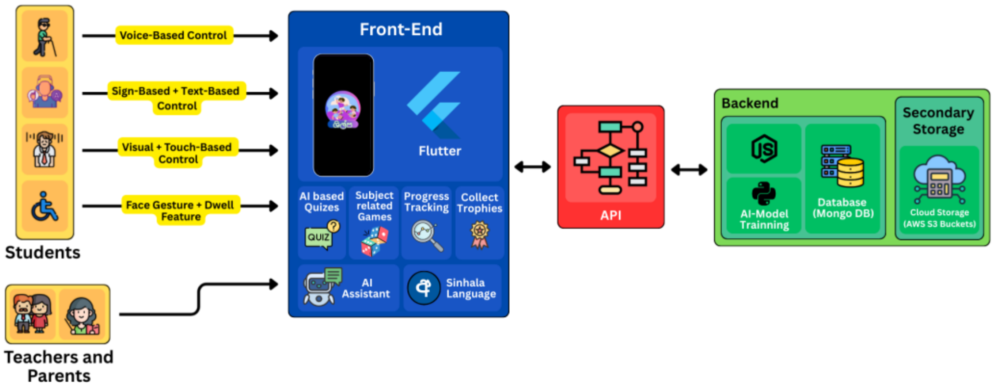

Figure 1: Shilpa system overview.

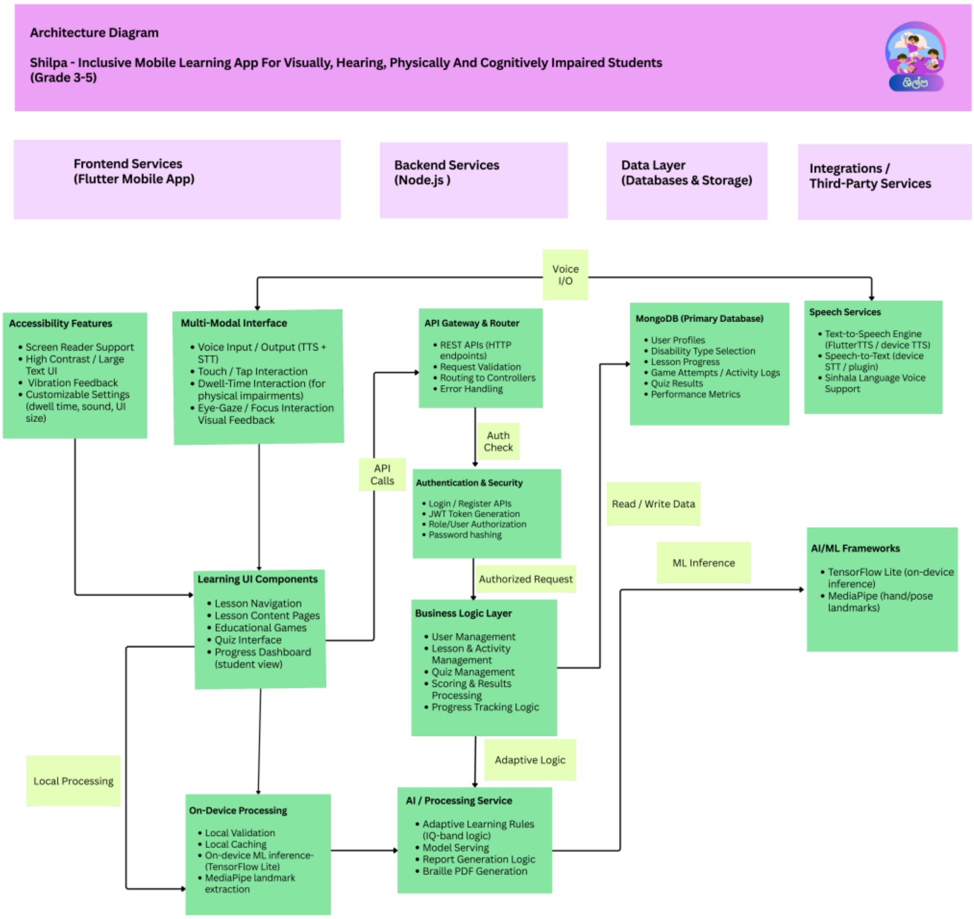

Figure 2: Shilpa system architecture diagram.

2.9 Project Requirements

2.9.1 Functional requirements

- Voice-based navigation, lesson delivery, and STT-based answer entry for visually impaired students; Braille-ready PDF generation, image upload, and automated CNN-based answer recognition.
- SSL video lessons with Sinhala subtitles, real-time MediaPipe + BiLSTM gesture recognition, sign-imitation games with gesture scoring, AI-generated post-lesson quizzes with badges, and a Sinhala chatbot for hearing-impaired students.
- Adaptive dwell-time selection, head-gaze tracking with blink-to-select, voice-command navigation in Sinhala and English, and a hybrid voice-plus-gaze interaction mode for physically impaired students.
- Game-based cognitive assessment, on-device TensorFlow Lite classification into below/average/above curriculum levels, and adaptive activity recommendation for cognitively impaired students.
- Role-based authentication, student / teacher / parent dashboards, progress tracking, achievement and badge systems, and progress reports for parents and teachers across all four modules.

2.9.2 Non-functional requirements

- Performance: end-to-end gesture recognition under 3 s on mid-range Android devices; activity load under 2 s and 60 fps interaction; quiz API response under 1 s.
- Usability: child-friendly Sinhala-medium interface, large interactive elements, simplified navigation, and consistent feedback.
- Reliability: validation of voice and image inputs; graceful degradation when ML services or network are unavailable; controlled command vocabulary.
- Accessibility: WCAG 2.1 AA-aligned design, screen reader compatibility, voice control, and customisable dwell time and button size.
- Security: HTTPS / TLS 1.3+, JWT-based session management, hashed passwords, anonymised data storage, and least-privilege access.

- Maintainability: modular architecture, well-documented APIs, automated tests covering core features, and Git-based version control.

2.9.3 Other requirements

- Personalisation: per-user calibration for gaze, adaptive dwell-time, voice speed, audio clarity, and curriculum-level adaptation.
- Localisation: Sinhala-first interface, bilingual voice commands, and alignment with the Sri Lankan Grade 3–5 curriculum.
- Safety and error prevention: emergency stop commands, accidental-touch cancellation, visual countdowns, and stability filtering.
- Logging: continuous performance, recognition accuracy, and interaction logs for evaluation and continuous improvement.

2.10 Consideration of the Aspects of the System

2.10.1 Social aspects

Shilpa contributes to social inclusion by enabling students with diverse impairments to access digital learning independently. By providing voice, Braille, sign-language, multimodal-touch-free, and adaptive cognitive interactions in a single Sinhala-medium platform, the system reduces reliance on external assistance, promotes equal learning opportunities, narrows the digital divide, and supports inclusive participation in both school and home environments.

2.10.2 Security aspects

Authentication is implemented via Firebase Auth and JWT-based session tokens. Inputs are validated on both client and server (voice commands filtered through confidence thresholds and an allow-list, gaze inputs filtered for spikes and stability, image uploads validated for type and size). Data uploaded to the system, including answer-sheet images and interaction logs, is stored securely and used only for evaluation and system improvement. Communications use HTTPS / TLS, and stored personal data is anonymised.

2.10.3 Ethical aspects

The system is designed with fairness, accessibility, and user well-being in mind. Multiple interaction methods are provided so that no single ability is privileged; classification of students into curriculum levels is used only to deliver suitable activities, never as a label or limit on ability; consent was obtained from schools and guardians for all data collection sessions; and the system avoids causing fatigue or frustration through adaptive thresholds and child-friendly feedback. The system supplements — and never replaces — teachers and special-education professionals.

2.10.4 Limitations

- CNN-based Braille recognition accuracy depends on lighting, alignment, and writing consistency in the captured answer sheet image.
- BiLSTM SSL recognition accuracy depends on adequate ambient lighting and gestures performed within the camera frame; partial occlusion or rapid execution can cause errors.
- Gaze-based navigation depends on lighting and camera quality, and continuous use can cause eye fatigue; voice recognition is affected by background noise and a fixed command vocabulary.
- The cognitive classifier is trained on a dataset that mixes synthetic and limited real-world samples; real-world accuracy is therefore lower than synthetic-test accuracy and more real data is needed.
- Some features (notably the Sinhala chatbot and parts of the speech pipeline) require an internet connection.
- The current build targets Android only; iOS and tablet-optimised layouts are planned future work.

2.11 Commercialization Aspects of the Product

2.11.1 Target audience

- Sri Lankan primary school students in Grades 3–5 with visual, hearing, physical, or cognitive learning difficulties.
- Special education teachers and educators in inclusive classrooms.

- Parents and guardians supporting children's learning at home.
- Schools, special-education units, and government agencies such as the Ministry of Education and the National Institute of Education (NIE).
- Disability NGOs and inclusive-education programmes.

2.11.2 Market space

- Inclusive education technology (EdTech) sector with growing demand for accessible Sinhala-medium learning solutions.
- Increasing adoption of mobile-based learning platforms in primary education in Sri Lanka.
- A clear local gap in unified, multi-disability, curriculum-aligned platforms suitable for low-resource environments.
- Potential expansion to other regional sign languages (e.g., Tamil Sign Language) and to neighbouring South Asian markets.

2.11.3 Revenue earning

Revenue Model	Description	Target Customer
Freemium consumer app	Core lessons free; advanced curriculum levels, Braille assessment analytics, offline mode, and detailed progress dashboards through a monthly subscription	Individual families (LKR 250–500/month)
Institutional licence	School-wide or district-level annual licence with centralised teacher admin dashboard	Government deaf and special schools, Provincial Education Departments
Government / NGO grant	Partnership-based deployment with costs covered through education or disability grants	Ministry of Education, UNICEF, WHO, local NGOs
Institutional content	Paid additional Sinhala-medium lessons, Braille assessment packs, and gamified curriculum modules	Schools and Provincial Education Departments
White-label SaaS	Platform adapted for other regional sign languages (e.g., Tamil)	Regional EdTech partners and international NGOs

Table 1: Proposed revenue models for the integrated Shilpa platform.

2.12 Testing and Implementation

Implementation followed a three-tier solution architecture (Flutter mobile client → Node.js / Express backend gateway → Python FastAPI ML service), structured across iterative Agile sprints. Testing was performed across four levels — unit, integration, system, and user acceptance — and module-specific testing strategies are summarised below. Detailed test results are reported in Chapter 3.

2.12.1 Visual impairment — implementation and testing

The visual module integrates Flutter-based voice navigation with a CNN Braille recogniser. Implementation includes the mobile UI, the Braille-ready PDF generator, and the image-upload pipeline. Testing covered model training and validation accuracy, full-pipeline behaviour from PDF generation through OpenCV preprocessing, per-cell segmentation, CNN prediction, answer matching, and audio-feedback delivery, and usability with visually impaired students at the partner schools.

2.12.2 Hearing impairment — implementation and testing

The hearing module integrates SSL video playback, MediaPipe-based landmark capture, and the BiLSTM gesture recogniser deployed via TensorFlow Lite. Testing included unit tests for backend controllers and JWT validation, integration tests across the Flutter → Node → Python request chain, system tests for end-to-end flows including video upload and recognition, and field user acceptance testing with twelve Grade 3–5 students at the School for the Deaf in Tangalle, with structured observation and feedback from special-education teachers.

2.12.3 Physical impairment — implementation and testing

The physical module is implemented entirely in Flutter using service-oriented singletons (AdaptiveDwellService, EyeTrackingService, SpeechService, VoiceCommandParser, VoiceFocusService, InteractionStatusService) and the polymorphic InputAwareButton. Testing involved five primary-school students aged 7–10 in classroom conditions, with each session lasting 25–30 minutes, using a Samsung smartphone with a 13 MP front camera. Four interaction modes were evaluated against task completion time, selection accuracy, error rate, false-activation rate, and qualitative feedback.

2.12.4 Cognitive impairment — implementation and testing

The cognitive module uses a Flutter frontend, Node.js backend, MongoDB storage, and an on-device TensorFlow Lite classifier. Testing included functional tests of metric calculation, integration tests of the API and data-storage pipeline, model evaluation on a synthetic test set with precision, recall, F1, and a confusion matrix, evaluation on a real student dataset, custom test-case analysis (including extreme low-performance scenarios), and a pilot survey with students, teachers, and parents at the partner schools.

3. RESULTS AND DISCUSSION

3.1 Results

The four modules of Shilpa were evaluated independently and then in combination as a unified mobile platform. The per-module results are summarised below.

3.1.1 Visual impairment results

The Braille recognition CNN was trained on the manually collected Braille digit dataset after preprocessing (resizing, thresholding, and per-cell segmentation). Training accuracy reached approximately 92 % while validation accuracy stabilised between 88 % and 89 %, indicating good generalisation and minimal overfitting. The trained model was further tested with real answer-sheet images captured through the mobile application: recognition was reliable when images were captured under good lighting with proper alignment and clearly written Braille dots, and minor errors occurred when image quality was low or writing was unclear. The complete workflow — voice navigation, Braille-ready PDF generation, answer-sheet upload, CNN prediction, automated evaluation, and audio feedback — operated successfully end-to-end. Usability testing showed that visually impaired students could navigate lessons, complete quizzes, and receive feedback with minimal external assistance.

3.1.2 Hearing impairment results

The BiLSTM SSL recogniser reached approximately 97–98 % training accuracy and stabilised at approximately 96–97 % validation accuracy with a small generalisation gap, confirming that the BiLSTM architecture effectively learned the temporal relationships in the hand-landmark sequences required for dynamic SSL recognition. Field testing with twelve Grade 3–5 students at the School for the Deaf in Tangalle showed strong engagement with sign-imitation games where camera feedback was visible during the gesture; students repeated signs voluntarily to improve their score, indicating a self-directed practice loop. Per-user calibration during the first session reduced false negatives noticeably for students with smaller hands or shorter range of motion. End-to-end measured performance is summarised in Table 3.1.

Metric	Measured Value
Average gesture recognition latency	2.3 s
Lesson video start time (cached)	0.8 s
Quiz generation API response time	0.6 s
Cold-start time of Python ML service	12 s
Mobile app cold-start time	1.9 s

Table 3.1: Hearing impairment module — measured system performance.

3.1.3 Physical impairment results

The four interaction modes were each evaluated independently. The adaptive dwell-time mechanism reduced average task completion time from 17.6 s (fixed dwell) to 14.2 s (adaptive), improved selection accuracy from 81 % to 89 %, and reduced error rate from 19 % to 11 % and false-activation rate from 14 % to 7 %. Gaze-only navigation showed 79 % selection accuracy with calibration success in 5/5 sessions and an average cursor stability of 84 %. Voice-only interaction achieved 91 % command-recognition accuracy with the lowest false-activation count among single modes. The hybrid voice-plus-gaze mode achieved the best overall performance with 10.3 s task completion time, 94 % selection accuracy, 6 % error rate, and 3 % false-activation rate. The full per-mode comparison is shown in Table 3.2.

Interaction Mode	Completion Time (s)	Selection Accuracy (%)	Error Rate (%)	False Activation Rate (%)
Voice Only	12.7	91	9	5
Gaze Only	18.4	79	21	16
Gaze + Adaptive Dwell	14.2	89	11	7
Hybrid (Voice + Gaze)	10.3	94	6	3

Table 3.2: Physical impairment module — comparison of interaction modes.

3.1.4 Cognitive impairment results

On synthetic test data the cognitive classifier reached near-perfect accuracy across the three curriculum levels (below average, average, above average). On real student data collected from 23 students the model reached approximately 82–83 % overall classification accuracy, indicating strong capacity to assess student performance levels in practical classroom scenarios. Misclassifications were observed mainly between average and above-average classes, consistent with the natural overlap of behavioural metrics for high performers. Functional, integration, and model tests all passed (T-01 metric calculation, T-02 API submission, T-03 classification — all status Pass), and the pilot survey with students, teachers, and parents reported strong engagement, easy-to-understand activities, and useful personalised recommendations.

3.2 Research Findings

Across the four modules, the evaluation supports the central thesis of the project: integrating multiple accessibility modules within a single Sinhala-medium platform enhances usability, supports independent learning, and reduces accessibility barriers for inclusive primary education in the Sri Lankan context.

- Image-based Braille recognition with a CNN delivers accuracy sufficient for automated assessment in real classroom conditions, and the Braille-based assessment workflow successfully bridges traditional Braille writing practice with digital evaluation.
- BiLSTM with MediaPipe landmarks is well suited to the temporal characteristics of SSL gestures, achieves high accuracy with a small generalisation gap, and runs on mid-range Android devices with sub-3-second end-to-end recognition latency.
- Combining gaze, voice, and adaptive dwell-time into a hybrid voice-plus-gaze fusion mode produces the best multimodal performance — the lowest task time and the highest selection accuracy — and per-user calibration is necessary for fair gaze-based recognition across hand sizes and head-movement profiles.
- On-device TensorFlow Lite classification of curriculum level is feasible from gamified-assessment behavioural metrics; real-world accuracy is meaningful but lower than synthetic-test accuracy, indicating the value of larger real datasets.
- Across all four modules, integrating multiple feedback channels — audio, vibration, visual, and gamified feedback — improved engagement and self-directed learning behaviour.

3.3 Discussion

Compared to systems reviewed in Chapter 1, Shilpa is unique in combining all four impairment modules in one Sinhala-medium platform aligned with the Sri Lankan

Grade 3–5 syllabus. Where prior visual-impairment work focuses on either reading aids or Braille recognition in isolation, Shilpa unifies voice navigation, Braille-ready PDF generation, and CNN-based answer recognition into a single learning workflow. Where prior hearing-impairment work focuses on translation or static-image recognition with ASL or BSL, Shilpa delivers SSL lessons with measurable real-time gesture scoring and curriculum alignment. Where prior physical-impairment work validates each modality in isolation, Shilpa fuses them in a hybrid mode and adds child-specific safety controls. Where prior cognitive-impairment work targets English-speaking contexts, Shilpa delivers Sinhala-language adaptive content with on-device classification.

The results also reveal common limitations across modules. Recognition-based modules (CNN Braille, BiLSTM SSL, gaze tracking) depend on adequate lighting and image quality. Voice modules are sensitive to background noise and use predefined command vocabularies. The cognitive classifier would benefit from larger real-world datasets. These limitations point directly to the future-work plan in Chapter 4.

3.4 Summary of Each Student's Contribution

3.4.1 Wijesooriya C.D. — Visual Impairment Support (IT22078728)

- Designed and implemented the voice-driven Flutter interface with TTS, STT, vibration feedback, and screen-reader-compatible navigation.
- Designed and implemented the Braille-ready PDF generator and the answer-sheet image-upload pipeline.
- Collected the manual Braille-digit dataset and trained, evaluated, and deployed the Convolutional Neural Network (CNN) Braille recognition model used for automated quiz evaluation.
- Implemented OpenCV preprocessing, per-cell segmentation, and prediction feedback delivery via audio.

3.4.2 Thanthirige K.S.G.I. — Hearing Impairment Support (IT22177650)

- Designed and delivered curriculum-aligned SSL video lessons with Sinhala subtitles for Grade 3–5 learners.
- Constructed the SSL dataset (word, letter, and number signs) collected with the support of the School for the Deaf in Tangalle.
- Implemented the MediaPipe-based hand-landmark extraction pipeline used to convert raw video into structured sequence input.
- Designed, trained, and deployed the BiLSTM gesture recognition model to TensorFlow Lite, including dropout regularisation and a SoftMax probability layer.
- Developed the sign-imitation games, instant-feedback mechanism, post-lesson quiz module, and the progress storage and reporting features for teachers and parents.
- Coordinated the field evaluation at the School for the Deaf in Tangalle.

3.4.3 Perera G.W.R.S. — Physical Impairment Support (IT22064622)

- Designed and implemented the Adaptive Multimodal Interaction Optimisation Framework — including the AdaptiveDwellService, EyeTrackingService, SpeechService, VoiceCommandParser, VoiceFocusService, and InteractionStatusService — and the polymorphic InputAwareButton.
- Implemented the asymmetric adaptive dwell-time algorithm with SharedPreferences persistence and real-time UI feedback.
- Implemented head-gaze tracking with EMA smoothing, spike rejection, stability windows, 5-point calibration, and double-blink confirmation using Google ML Kit Face Detection.
- Implemented bilingual (Sinhala / English) speech recognition with Levenshtein-distance fuzzy matching, semantic-label fallback, and runtime registry-based voice focusing.
- Implemented the hybrid voice-plus-gaze fusion mode, evaluated four interaction modes with 5 students aged 7–10 in classroom conditions, and produced the per-mode comparison results.

3.4.4 Mathota Arachchi S.S. — Cognitive Impairment Support (IT22070876)

- Designed the closed-loop adaptive cognitive-support workflow combining game-based assessment, machine-learning classification, and personalised activity delivery.
- Implemented the three game-based assessment activities (shape matching, colour matching, bubble popping) and the behavioural metric collection.
- Engineered the 28-feature input vector and implemented the supervised TensorFlow / Keras neural classifier with batch normalisation, ReLU, dropout, and Softmax output across three CL classes.
- Trained the model with class weighting, exported to TensorFlow Lite, and deployed for on-device inference; integrated backend storage with the Node.js / MongoDB stack.
- Implemented the adaptive personalisation cycle that retrieves the latest CL prediction and matches activities to the student's level, and led the pilot evaluation with students, teachers, and parents.

4. CONCLUSION AND RECOMMENDATIONS

4.1 Conclusion

This research set out to address the absence of a single, locally relevant mobile learning platform that supports primary-school learners with visual, hearing, physical, or cognitive impairments in Sri Lanka. Shilpa demonstrates that this is technically achievable on commodity Android hardware. Within a single Flutter application, the platform delivers voice-driven learning and Braille-based automated assessment for visually impaired students, SSL video lessons with real-time BiLSTM-based gesture recognition and gamified imitation activities for hearing-impaired students, an Adaptive Multimodal Interaction Optimisation Framework (adaptive dwell time, head-gaze tracking with blink-to-select, voice control, and hybrid voice-plus-gaze fusion) for physically impaired students, and an on-device TensorFlow-Lite cognitive classifier with adaptive personalisation for students with cognitive and learning difficulties.

Across the four modules, the evaluation results confirmed that Shilpa meets its objectives. The CNN Braille recogniser achieved 88–92 % accuracy with reliable end-to-end answer-sheet evaluation; the BiLSTM SSL recogniser reached 96–98 % validation accuracy with sub-3-second end-to-end latency on mid-range Android devices and produced a measurable self-directed practice loop in field testing; the hybrid voice-plus-gaze interaction mode achieved 94 % selection accuracy with the lowest task time among modes; and the cognitive classifier reached 82–83 % accuracy on real student data with a working closed-loop adaptive cycle. Field evaluations at the National Institute of Education, the School for the Deaf in Tangalle, and partner primary schools confirmed that the system is usable by Grade 3–5 students with minimal external assistance.

4.2 Recommendations and Future Work

- Expand the Braille and SSL datasets through partnerships with additional schools to improve generalisation, especially for visually similar patterns and signs.

- Add image-enhancement preprocessing in both the Braille and SSL pipelines to mitigate the impact of low-light classroom conditions on landmark and dot detection stability.
- Move ML inference fully on-device using TensorFlow Lite for all modules, removing the dependency on the Python service for users without consistent internet connectivity, and replace cloud-based chatbot components with smaller distilled Sinhala models.
- Extend coverage from current digit/number content to letters and full subject vocabulary across the Grade 3–5 curriculum.
- Add iOS support and tablet-optimised layouts to broaden device coverage.
- Collect a larger longitudinal real-world dataset for the cognitive classifier and conduct an academic-term-long evaluation across multiple schools to measure learning gains rather than only short usability sessions.
- Pursue the four-phase scale-up roadmap: pilot at three Southern Province schools, national roll-out in partnership with the Ministry of Education, extension to inclusive mainstream schools and consumer launch, and regional adaptation for Tamil Sign Language and international NGO partnerships.

4.3 Concluding Remarks

Shilpa contributes to the advancement of inclusive education technology in three ways. First, it demonstrates that a single Flutter-based mobile application can integrate four distinct accessibility pipelines — Braille recognition, SSL gesture recognition, multimodal touch-free interaction, and cognitive curriculum-level classification — without specialised hardware. Second, it produces locally relevant artefacts: a curriculum-aligned SSL gesture dataset collected at a Sri Lankan deaf school, a Braille-ready PDF and answer-sheet evaluation workflow aligned with NIE Braille Press conventions, a per-user calibration step that improves recognition fairness, and an on-device adaptive cognitive classifier trained for Sinhala-medium learners. Third, it demonstrates through field evaluation that integrating multiple accessibility methods in one Sinhala-medium platform aligned with the Sri Lankan

Grade 3–5 syllabus increases independence, engagement, and learning outcomes for students with diverse needs.

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APPENDICES

Appendix A: Shilpa Mobile Application Logo

The official logo and colour palette used across the four accessibility modules of the application.

Appendix B: Application Loading and Landing Page Interface

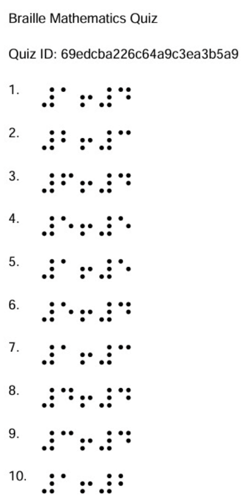
Screenshots of the application loading sequence, landing page, and top-level navigation.

Appendix C: User Interface for Disability Type Selection

Screens used at registration and login to select the disability category and route the learner to the corresponding accessibility module.

Appendix D: Visual Impairment — Audio-Based Math Quiz and Braille Quiz Generation

Audio-based math quiz screens for visually impaired learners and Braille-ready PDF generation, answer-sheet upload, and CNN-based evaluation result screens.

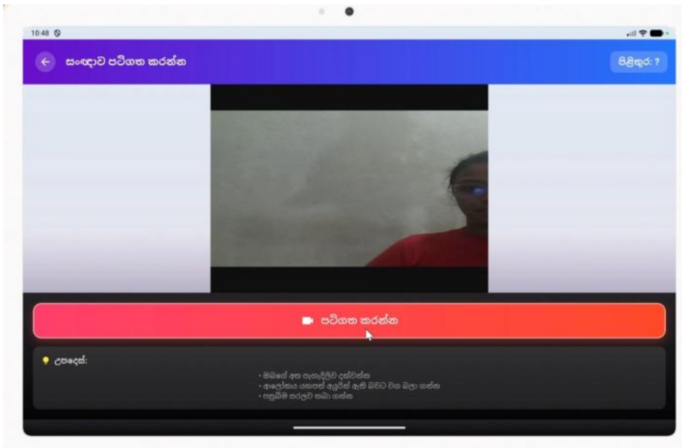

Braille Mathematics Quiz

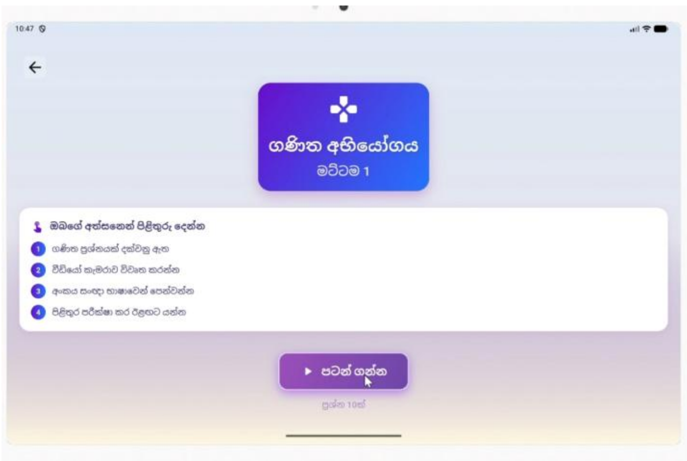
Quiz ID: 69edcba226c64a9c3ea3b5a9

1. ⠼⠁⠖⠼⠙
2. ⠼⠃⠖⠼⠉
3. ⠼⠋⠖⠼⠙
4. ⠼⠑⠖⠼⠑
5. ⠼⠁⠖⠼⠑
6. ⠼⠑⠖⠼⠙
7. ⠼⠁⠖⠼⠉
8. ⠼⠙⠖⠼⠙
9. ⠼⠉⠖⠼⠙
10. ⠼⠁⠖⠼⠃

Appendix E: Hearing Impairment — SSL Lesson Delivery and Sign Imitation Games

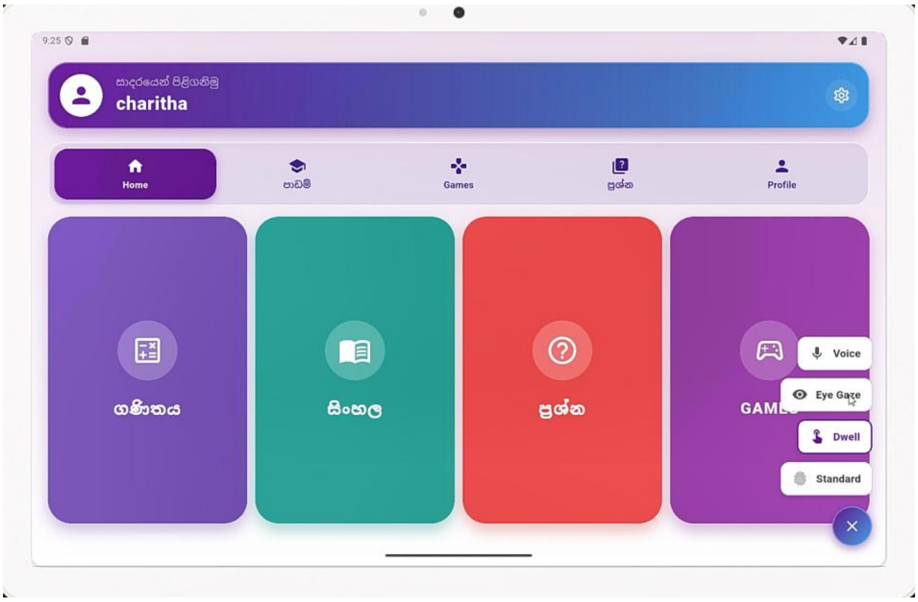
SSL video lesson playback with Sinhala subtitles, sign-imitation games with real-time gesture scoring, and game completion/reward displays.





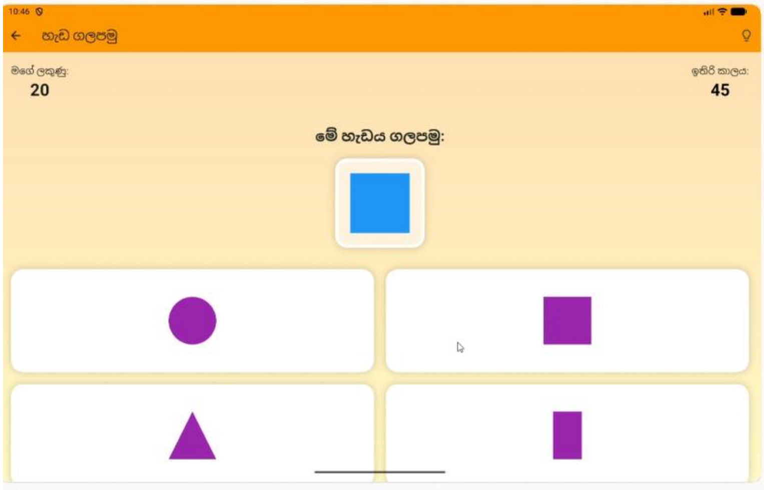
Appendix F: Physical Impairment — Multimodal Interaction Demonstration Screens

Adaptive dwell-time visual progress indicator, head-gaze cursor with stability and blink-to-select feedback, voice-command examples, and the hybrid voice-plus-gaze interaction screen.



Appendix G: Cognitive Impairment — Adaptive Game-Based Assessment Screens

Shape-matching, colour-matching, and bubble-popping assessment screens, the on-device curriculum-level prediction result, and personalized activity recommendations.



Appendix H: Data Collection at the National Institute of Education (NIE)

Photographs and notes from data collection sessions at the NIE Maharagama and the NIE Braille Press.